



8496 - Searching for New Formation of the Earliest Solar System Materials in the Infant RW Aur A Exoplanetary System

Cycle: 4, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

We propose to use archival MIRI/MRS spectral measurements of the RW Aur A infant evolving solar system, taken under PID 1282 (PI Henning) in early 2023 and due to become public on 16 Oct 2024. Our goal is to search for newly created calcium-aluminum inclusions (CAIs), and chondrules in the aftermath of the major planetesimal disruption event in RW Aur A's inner accretion disk seen in 2014-2020. We will do this by comparing the archival MRS spectrum to archival Spitzer/IRS spectral data of the system taken by Furlan et al. in 2007, and searching for predicted new emission features in the 7-21 um range due to glassy silicates and alumina oxides.

OBSERVING DESCRIPTION